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INTERIM FEDERAL SPECIFICATION

SUCTION AND PRESSURE APPARATUS, SURGICAL, EXPLOSION-PROOF, SINGLE COMPRESSOR, MOBILE, 110 VOLT, 60 CYCLE, AC: (FOR EAR, NOSE AND THROAT PROCEDURES)

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification GG-S-1090.

1. SCOPE

1.1 This specification covers a suction and pressure unit, suitable for ear, nose and throat procedures in Class I, Group C, explosive atmospheres.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

FSC 6515

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SPECIFICATIONS

Federal

CC-M-636	Motor, Fractional Horsepower (Alternating Current).
FF-C-77	Casters; Institutional Duty.
QQ-S-698	Steel, Sheet and Strip, Low-Carbon.
TT-E-491	Enamel; Gloss, Synthetic (for Metal and Wood Hospital Furniture).
TT-P-659	Primer-Surfacer; Synthetic, Tints and White, (for Metal and Wood Surfaces).
UU-P-271	Paper, Wrapping, Waterproofed Kraft.
ZZ-S-751	Stoppers, Bottle, Rubber.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.

Military

MIL-P-16232	Phosphate Coatings, Heavy, Manganese or Zinc Base (for Ferrous Metals).

STANDARDS

Federal

FED-STD-66	Steel; Chemical Composition and Hardenability.
FED-STD-102	Preservation, Packaging, and Packing Levels.
FED-STD-123	Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. 151	Metals; Test Methods.
FED-STD-595	Colors.
Fed. Test Method Std. 601	Rubber; Sampling and Testing.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

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(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards, as outlined under General Information in the Index of Federal Specifications and Standards, and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20325).

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.)

(Federal Government activities may obtain copies of Federal Specifications and Standards, and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

(Copies of Military specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

AMERICAN SOCIETY FOR TESTING MATERIALS

ASTM Designation C147	Method of Internal Pressure Test on Glass Containers.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION

NFPA 56A	Standard for the use of Inhalation Anesthetics, 1970.

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, Mass. 02110.)

UNDERWRITERS LABORATORIES, INC. STANDARDS

UL674(c)	Standard for Electric Motors and Generators for Use in Hazardous Locations, Class I, Group C and D. Part II - Fractional Horsepower Motors.
UL894	Standard for Snap Switches for Use in Hazardous Locations.

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(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 1285 Walt Whitman Road, Melville, Long Island, New York, 11746; 207 East Ohio Street, Chicago, Illinois 60611; or 1655 Scott Boulevard, Santa Clara, California, 95050.)

3. REQUIREMENTS

3.1 Fire and casualty hazards.

3.1.1 The contractor shall submit proof, prior to delivery, that the items to be delivered under the contract conform to the applicable requirements of UL-674(c), 894, 62 including appendices thereto, and NFPA 56A. Proof shall consist of one of the following:

A. UL label or UL listing of the product, provided the contractor agrees to have Underwriters' Laboratories, Inc. forward a copy of the UL report directly to the Contracting Officer, upon the latter's request.

B. Laboratory report from a qualified independent laboratory. If an independent laboratory is used, it must be adequately equipped and competent to determine compliance with UL standards. The qualification of such laboratory shall be determined by the Government which reserves the right to require a facilities report and to conduct a facilities survey. Any laboratory report submitted shall include actual test methods, data taken, and results of tests. Detailed findings of examinations necessary to determine compliance shall also be included. The test report shall also have a section that describes the test sample in such a way that the report can be used to insure identity of production items with the test sample. This section shall be supplemented with construction information such as materials used, spacings between electrical components, types of bushings, photographs, wiring diagrams, sketches, and parts list, all appropriately identified. Compliance with this paragraph does not excuse compliance with all other requirements of the contract and specifications.

3.2 Material.

3.2.1 Carbon steel. Carbon steel shall conform to QQ-S-698 for C1008, C1010, or C1020 commercial-quality, cold-rolled sheet, with a No. 1 (matte) finish. Commercial tolerances shall apply to all gages specified hereinafter.

3.2.2 Corrosion-resisting steel. The corrosion-resisting steel shall conform to AISI 304 of FED-STD-66.

3.2.3 Glass. The glass shall be clear and annealed.

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3.2.4 Rubber tubing. Rubber tubing shall be electrically conductive and shall have an inside diameter of 1/4 inch with a 1/8 inch wall. The tubing shall have the following physical properties:

Tensile strength (pounds per square inch, minimum)	1,200
Ultimate elongation (percent, minimum)	500
Tensile strength (psi minimum) after aging	900

The tubing shall be electrically conductive, and shall conform to the requirements for Reduction of Electrostatic Hazard-Conductive Accessories of NFPA 56A, when tested as specified in 4.4.1.2.2. In addition, the electrical resistance of the tubing shall be measured after exposure to oxygen aging, bending and flexing, and oil, as specified in 4.4.1.2.3. Any value in excess of 1 megohm shall be cause for rejection. Proof of acceptability of tubing shall consist of either a listing of the apparatus by Underwriters' Laboratories, Inc., or a laboratory report from a qualified laboratory. (see 3.1.1.B)

3.2.5 Rubber stoppers. The stoppers shall conform to ZZ-S-751, type B, Grade I.

3.2.6 Primer surfacer. The primer surfacer shall conform to the requirements of TT-P-659.

3.2.7 Enamel. The enamel shall conform to class B of TT-E-491, except that phthalocyanine pigments may be used to obtain the specified color. The color of the enamel shall be light green, conforming to color 14241 of FED-STD-595.

3.3 Design and construction. The suction and pressure apparatus shall consist of a minimum 1/6 horsepower, explosion-proof, motor-driven, rotary compressor; suction and pressure system; and filter systems. The apparatus shall be properly connected, provided with the required accessories (see 3.4), and compactly mounted in a well-built, movable cabinet.

3.3.1 Cabinet body. The cabinet shall be a drawer and cupboard type, constructed of carbon steel 0.050 inch, No. 18 United States Standard (U.S.S.) gage. The body shall be constructed of one continuous piece of carbon steel, and shall be 20 inches long, 16 inches wide and 32 inches high, including the supporting legs and casters. Tolerance on all dimensions shall be $\pm$ 1/4 inch.

3.3.1.1 Top. The top shall be constructed of corrosion-resisting steel 0.031 inch (No. 22, U.S.S. gage) and shall be 20 inches long by 16 inches wide. Tolerance shall be as specified in 3.3.1. All sides shall have a full-hemmed, 1 inch $\pm$ 1/8 inch flange. The corners shall be continuously welded and ground smooth. The top shall be closely fitted over a subtop, constructed of carbon steel not less than 0.050 inch (No. 18 U.S.S. gage) with 1-inch apron, and shall be closely and securely spot-welded to the cabinet body.

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3.3.1.2 Drawers. The cabinet shall be fitted with three drawers constructed of carbon steel. Directly below the cabinet top shall be centrally located a horizontal bank of two drawers, each 5 inches wide, 3-1/4 inches high, and 14 inches long. Immediately below these shall be a full-width drawer, 14 inches wide, 4 inches high, and 14 inches long. Tolerance on dimensions shall be ± 1/8 inch. The drawer bodies shall be not less than 0.031 inch (No. 22, U.S.S. gage) with single-panel drawer fronts not less than 0.078 inch (No. 14, U.S.S. gage) spot-welded to the drawer bodies. The drawers shall be fitted with channel suspensions not less than 0.050 inch (No. 18, U.S.S. gage) with suitable drawer stops to prevent accidental withdrawal from the housing. Channel suspensions shall telescope into and ride smoothly, but not loosely in channel runner properly located and welded to the cabinet housing. Each drawer pull shall be the single-knob type made of corrosion-resisting steel, or chromium over nickel, plated steel. As an alternate to the single panel front specified above, the drawer fronts shall be of double-wall construction. The drawer bodies and fronts shall be designed and constructed so that when closed, the drawer fronts shall be flush with the front of the cabinet body.

3.3.1.3 Compartment. Directly below the large drawer shall be a sound-deadened compartment 20 inches wide, 16 inches long and 12 inches high. Tolerance shall be ± 1/4 inch. The bottom of the compartment shall be constructed of carbon steel not less than 0.0625 inch (No. 16, U.S.S. gage), properly formed and reinforced, closely spot-welded to the cabinet body and suitably perforated for proper ventilation of the motor and compressor unit. Two angle-iron members, each 1-1/4 by 1-1/4 by 1/8 by 12 inches long, shall be welded to the bottom. These shall be centrally located, running parallel from front to back in the compartment, forming a channeled section for securing the motor and compressor unit in an easily-accessible manner. As an alternate to the motor mounting specified above, the motor shall be mounted on a wooden base 8-1/2 ± 1/4 inches by 10 ± 1/4 inches by 1-5/16 ± 1/8 inches, which shall be dovetailed and fit in a parallel channeled plate with a back stop. As an alternate to the above, the motor shall be bolted to the angle irons, and the angle irons mounted on vibration mountings which shall be bolted or riveted to the bottom. The plate shall be fabricated of 0.037 inch (No. 20, U.S.S. revised gage) carbon steel, and shall be spotwelded to the compartment floor. The wooden base shall be secured to the compartment floor by means of a 1-5/16 inch hexagonal head bolt. The door front shall be of double-wall construction. The door shall be so designed and constructed that when closed, the door front shall be flush with the front of the cabinet.

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3.3.1.4 Guard rail. A guard rail, constructed of not less than 3/8 inch corrosion-resisting steel, shall extend continuously around the ends and back of the cabinet top parallel to and extending 2 inches above the plane of the cabinet. The rail shall be supported by uprights of similar material at each corner and two evenly-spaced positions at the rear. The rail and top shall be secured to the cabinet and subtop by means of machine screws, lockwashers, and nuts. The rail shall be equipped with 12 springclips, for holding twelve 2-ounce, glass-stoppered, glass salt bottles. The springclips shall be constructed of corrosion-resisting steel 0.031 inch (No. 22 U.S.S. gage) to 0.037 inch, No. 20, U.S.S. gage, by 1/2 inch wide, properly formed to fit and secure the bottles to the rail.

3.3.1.5 Legs and casters. Each cabinet shall be supported on four legs of suitable design and size to which shall be attached electrically conductive rubber casters conforming to FF-C-77 for type II, class A, not less than size 3 (4-inch diameter wheel shall be acceptable provided it meets minimum requirements for load rating for size 3), style b, 7/16-inch diameter, 1-1/2 inch long stem with retaining band. Two of the casters shall have a locking device. The cabinet with casters in place shall conform to the requirements for Reduction of Electrostatic Hazard of NFPA 56A, when tested as specified in 4.4.3.1.

3.3.2 Motor. The motor shall conform to CC-M-636 for type I, class 1b, style B4, continuous duty, 1725 revolutions-per-minute (r.p.m.), 50° Centigrade (C.) ambient. The motor shall have not less than 1/6 horsepower and shall operate on 115 volts, 60 cycles, alternating current. Proof of acceptability of the motor shall consist of either a listing by Underwriters Laboratories, Inc., or a laboratory report (see 3.1). The motor shall be securely mounted within the cabinet, to eliminate vibration and noise. The motor shall be operated by an easily-accessible, enclosed-type, explosion-proof switch, plainly marked to indicate the off-and-on positions. The switch shall be mounted on the side or front of the cabinet near the top, and shall have a Class 1, Group C rating in accordance with Underwriters Laboratories Standard UL674(c). The motor shall be provided with a thermal overload protection and both the motor and switch shall be explosion-proof in accordance with UL674(c).

3.3.2.1 Electric wiring. Electrical plugs and receptacles within the unit shall be explosion-proof, approved for use in Class I, Group C locations. The unit shall be equipped with a flexible, insulated, three-wire cord, not less than 10 feet long. The third or ground wire shall be properly identified. The cord shall not be furnished with a male plug, but shall bear a red warning tag at the free end cautioning that only an explosion-proof attachment plug (to mate with the hospital explosion-proof receptacle) shall be installed by the using activity. The unit shall withstand the dielectric strength test specified in 4.4.3.2.

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3.3.2.2 Conductivity. When tested as specified in 4.4.3.1, electrical conductivity of the apparatus shall be such that the resistance between a metallic object placed on the cabinet and a metallic plate placed under any caster shall be not more than 250,000 ohms when the rest of the unit is insulated from the floor.

3.3.3 Compressor. The compressor shall be a rotary, high-pressure type, connected directly to the motor shaft or attached by a strong flexible coupling, to insure quiet and even operation. No springs, valves, or belts shall be used. The compressor shall produce both positive and negative pressure which shall be controlled by separate regulating devices in a manner to permit the simultaneous use of both pressures if desired. The capacity of the compressor shall be not less than 1/2 cubic foot of air per minute at the operative ends of the tubing, at a motor speed of 1,725 revolutions per minute.

3.3.3.1 Positive-pressure system. The compressor shall develop, and transmit by suitable rubber tubing, a positive air pressure of not less than 20 pounds per square inch. A control valve on the outlet shall permit regulation of the air pressure from zero to maximum. A gage shall be mounted on the outside of the cabinet on the positive-pressure outlet to measure the positive pressure from zero to 30 psi in increments of one psi. The gage shall have an accuracy of ± 2 percent in the middle half of the scale, and ± 3 percent in the remainder.

3.3.3.2 Negative-pressure system. The compressor shall develop, and transmit by suitable rubber tubing, a negative pressure of not less than 20 inches of mercury in the collection bottle, easily variable to zero value by means of regulating valve. A gage shall be mounted on the outside of the cabinet, securely fastened to the negative-pressure outlet to show all values of the negative pressure from zero to 30 inches mercury, in increments of one-inch, accurately and plainly. A device shall be provided which shall cut off the negative pressure from the suction bottle to prevent overflow. The gage shall have an accuracy of ± 2 percent in the middle half of the scale, and ± 3 percent in the remainder.

3.3.3.2.1 Suction bottle. A widemouthed 32-ounce capacity suction bottle shall be securely mounted on the side of the cabinet in a way that permits easy removal for cleaning purposes. The bottle shall be connected to the negative pressure system by couplings and rubber tubing attached to a rubber closure. The bottle shall withstand a vacuum of 25 inches minimum of mercury for one minute without failure, when tested as specified in 4.4.2.1. Suction bottle shall also pass the test in 4.4.3.4.

3.3.3.3 Filter systems. Two enclosed filters shall be provided, one in the positive-pressure and one in the negative-pressure system, to insure adequate air purification. The filters shall be easily replaceable. The pressure and vacuum filter jars shall pass the test specified in 4.4.3.4.

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3.4 Accessories. Each cabinet shall be supplied with the following accessories:

(a) One rubber spray tube with screw-type metal couplings, and a tapered thumb-cutoff to fit standard atomizer spray bottles. The couplings shall be fabricated of chromium-plated brass or steel and shall be a taper-slip type with positive screw locks. The male portion of the coupling shall be tapered from a diameter of 0.351 ± 0.005 inch decreasing for a distance of 0.250 or 0.275, ± 0.005 inch to a diameter of 0.250 ± 0.005 inch. The female portion of the coupling shall have a corresponding taper. The male part of the coupling shall be provided with a threaded portion (7/16-inch by 27-NC thread) which shall mate with a threaded collar on the female part of the coupling. The collar and threaded portion shall insure locking of the coupling to form a tight leakproof connection.

(b) Twelve 2-ounce glass-stoppered salt-mouth glass bottles.

(c) Sufficient rubber tubing to perform its intended function. The tubing shall be 1/4 ± 1/16 inch in inside diameter with 1/8 ± 1/64 inch wall, and shall be fabricated of electrically conductive rubber. The tubing shall be provided with couplings fabricated of brass or steel which shall be chromium-plated over nickel. The couplings shall be of a taper-slip type with positive screw locks. Couplings shall be as specified in (a) above.

(d) One Yankauer laryngeal cannula, whose proximal end shall be fitted with a male taper slip connection to mate with the hose coupling on the suction tubing.

(e) One glass sinus-cleanser bottle, complete with one small and one large flexible cannula, curved nose piece, thumb cut-off, and metal adapter.

The accessories which are subject to constant use, including bottles and tube assemblies, shall be securely mounted on, or securely attached to, the cabinet in a manner to insure easy accessibility, (see 6.5).

3.5 Finish. All interior and exterior surfaces of the cabinet, except the top and the hardware, shall be cleaned, primed and enameled.

3.5.1 Cleaning and preparation. Before finishing, parts to be painted shall be cleaned with suitable solvents and phosphate treated in accordance with type M, class 1 of MIL-P-16232, or other equivalent chemicals to provide a satisfactory bond for the finishing materials. All castings shall be polished prior to surface treatment and application of primer.

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3.5.2 Primer. After cleaning and preparation, the parts shall be given one coat of a synthetic primer conforming to TT-P-659. The primer shall be baked on for not less than 45 minutes, at a temperature of 250° Fahrenheit (F.), or for an equivalent baking period at a higher or lower temperature as recommended by the paint manufacturer. The primer shall be applied so as to level out and produce a uniform surface, free of pebbles and wrinkles.

3.5.3 Enamel. After the primer has been allowed to cool, all surfaces shall be covered with two coats of synthetic gloss enamel conforming to class B of TT-E-491, except that phthalocyanine pigments may be used to obtain the specified color. Each coat shall be baked on separately for not less than 45 minutes at a temperature 250° F. or for an equivalent baking period at a higher or lower temperature as recommended by the paint manufacturer. The enamel coats shall be applied so as to level out and produce a final, smooth, uniform surface without runs, wrinkles, orangepeel, or grit, and without areas of thin or no film. The total dry-film thickness shall be not less than 0.002 inch at any point.

3.5.4 Cabinet top. The corrosion-resisting steel cabinet top shall be given a uniform No. 4 commercial finish.

3.5.5 Hardware. Hardware, including the fittings and fixtures, shall be chromium-plated over nickel, polished to a mirror finish; or corrosion-resisting steel, polished to a No. 4 commercial finish.

3.6 Identification marking. Each unit shall be permanently and legibly marked with the National Stock Number, name or registered trademark of the manufacturer, item identification, the contract or purchase order number, the letters U.S., the serial number, voltage, amperage, voltage, and frequency rating, oiling instructions, and such other data as may be required by Federal laws. In addition, the motor shall be marked in accordance with CC-M-636. Marking secured by adhesives shall not be used. Marking, except motor marking, shall be of a size to be clearly legible, and located to allow its being visible during use of the item.

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3.7 Service Data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the Contracting Officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual, for review and comment by the Directorate of Medical Material, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Material, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the Contractor shall furnish 15 copies of the approved technical manual to the Contracting Officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the Contracting Officer. Waiver of submission of 15 additional copies shall be at the option of the Government.

3.7.1 Center of gravity and net weight information. Unless otherwise specified, the contractor shall prepare a drawing, figure, or illustration, indicating the position of the vertical center of gravity of the unit, with reference to its base. The net weight of the unit shall be also indicated. Four (4) copies shall be submitted by the contractor directly to the contracting officer. In the event the required information has been submitted previously, no further report is necessary unless changes in the unit involve a variation greater than ten percent in center of gravity or net weight.

3.8 Welding. Welding shall be durable, sound in every detail and in accordance with latest practices recommended by the American Welding Society. All welds shall be produced by qualified welding operators with satisfactory welding equipment using satisfactory filler metal or electrode where applicable. Surfaces to be welded shall be free from grease, oil, oxides or any other foreign matter. Any cleaning method which will remove these substances and leave the surface clean is satisfactory. All aluminum parts shall be welded within 48 hours after a chemical cleaning treatment. Parts must be chemically recleaned if the 48 hour time period is exceeded. All weld joints shall blend into the adjacent metal in gradual, smooth curves. Beads shall be smooth and free of slag, excessive undercut, or excessive spatter. Sufficient metal shall be added to provide a suitable fillet or reinforcement unless otherwise specified. Fusion through the full thickness of the base metal, in areas other than those where 100% penetration is a requirement of joint design, are not acceptable when welding materials over 0.050 inch in thickness. Welds shall not be oxidized. All welds shall be continuous with no visible voids, pin holes, or cracks.

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3.9 Workmanship. Workmanship shall be first class throughout. The suction and pressure apparatus shall be free from any imperfections which affect its appearance or which may impair its serviceability. The finished apparatus shall operate smoothly and noiselessly, and shall efficiently perform its functions in a manner suitable for the purpose intended. The unit shall have no sharp edges.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Certificates of quality. Certificates of quality, supplied by the manufacturer of the raw materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number, the Item Identification, the name of the component or material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and Table I.

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TABLE I

	Inspection level	AQL (defects per hundred units)
For visual examination		
Major defects	II	1.0
Minor defects	II	4.0
For Dimensional Examination	S-3	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, as indicated in Table II. AQL shall be zero, except that, in the event of a failure during 100 percent inspection, the entire lot shall not be re-tested.

TABLE II

Component (unit of product)	Characteristic	Requirement	Test procedure	Inspection level
Metals	Chemical composition	3.2.1, 3.2.2	4.4.1.1.1	(1)
Rubber tubing	Physical properties	3.2.4	4.4.1.2.1	(2)
	Electrical conductivity	3.2.4	4.4.1.2.2	(2)
	Electrical resistance	3.2.4	4.4.1.2.3	(2)
Gage	Accuracy	3.3.3.1, 3.3.3.2	4.4.2.2	S-1
Suction bottle	Vacuum	3.3.3.2.1	4.4.2.1	S-1
End item	Electrical conductivity	3.3.2.2	4.4.3.1	100 percent
	Suction overflow	3.3.3.2	4.4.3.3	S-1
	Performance	3.3.3 to 3.3.3.3	4.4.3.4	100 percent
	Dielectric strength	3.3.2.1	4.4.3.2	100 percent

(1) Four ounces of each lot of metal used in the construction of the apparatus shall be selected for chemical analysis.

(2) Two samples of rubber tubing representative of each lot used in the manufacture of the apparatus shall be selected for test. Each sample shall be large enough to provide the required number of specimens for the tensile tests, both before and after aging, and for the electrical resistance tests. In addition, two 6-inch lengths of the intact tubing shall be taken, one for aging, and one as a control.

4.3 Examination. The suction and pressure apparatus shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

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Classification of defects*	
Major	
101	Any component or part missing.
102	Exposed edges and corners not free of burrs or sharpness.
103	Any weld not fused; not free of burns, cracks, or fractures.
104	Any threaded part stripped, or does not function smoothly.
105	Doors do not close flush, warped, or otherwise deformed.
106	Unit not clean; not free of dirt, oil, or grease.
107	Sound-deadning material missing.
108	Any bottle not free of cracks, breaks, chips, sharp or deformed areas.
109	Tapered slip-type couplings do not mate properly.
Minor	
201	Drawer does not operate smoothly.
202	Drawer does not close flush with cabinet.
203	Blistering, checking, or peeling of finish.
204	Bottle not stable; rocks on flat surface.
205	Any screw not drawn up tight; head not flush with the surface held.
206	Color of enamel finish not as specified.
207	Service data not included.
208	Identification marking missing, incomplete, illegible, or not properly attached.
209	Finish not free of tackiness, softness, or sags.
210	Finish not uniform; not free of imbedded foreign matter.
211	Finish not free of scratches, bare spots, or areas of visibly thin coating.

* Examination shall not be restricted to the classified possible defects tested above.

4.4 Tests. Tests shall be conducted, as necessary, to determine compliance with specification requirements.

4.4.1 Materials.

4.4.1.1 Steel.

4.4.1.1.1 Chemical composition. Tests for chemical analysis shall be conducted in accordance with method 111.1 or 112.1 of Federal Test Method Standard 151.

4.4.1.2 Rubber tubing.

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4.4.1.2.1 Physical properties. Physical properties of rubber tubing shall be determined in accordance with the appropriate test method of Federal Test Method Standard 601, as follows:

Property	Test method
Tensile strength	4111
Ultimate elongation	4121
Oxygen pressure	7111*
Air heat	7221*

* Only one of these tests is required.

4.4.1.2.2 Electrical conductivity. The rubber tubing shall be tested for electrical conductivity in accordance with the applicable procedure outlined in NFPA 56A.

4.4.1.2.3 Electrical resistance. The electrical resistance of the rubber tubing shall be determined after being subjected to the following tests.

4.4.1.2.3.1 Oxygen aging. Two 7-inch lengths of tubing shall be subjected to oxygen under pressure of 300 psi for a period of 96 hours at a temperature of 70° C. The electrical resistance of the tubing shall be measured before and after the oxygen aging.

4.4.1.2.3.2 Bending and flexing. Two 5-foot lengths of tubing, fitted with electrical test connections, shall be passed back and forth repeatedly for one hour; a minimum of 10 times per minute on a pulley 1-3/4 inches in diameter. The tubing shall be held in tension by a 5-pound weight suspended from the free end of the sample. The electrical resistance of the tubing shall be measured initially and at 20 minute intervals during the test, and at completion of test.

4.4.1.2.3.3 Oil. Two 7-inch lengths of tubing shall be submerged for 5 days in SAE 30 motor oil, type SA. The resistance of the tubing shall be measured before and after the test.

4.4.2 Components.

4.4.2.1 Vacuum (suction bottle). The suction bottle shall be subjected to an internal vacuum of 25 inches of mercury for a period of 1 minute without failure.

4.4.2.2 Gages. Gages shall be examined to determine that the calibrations conform to requirements. The gages shall be measured for accuracy using standard gages accurate to the nearest 1/2 percent.

4.4.3 End item.

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4.4.3.1 Electrical conductivity. The apparatus, complete with casters and rubber tubing, shall be tested in accordance with the applicable procedures outlined in NFPA 56A.

4.4.3.2 Dielectric strength. A dielectric test potential of 900 volts, 60 cycles, single-phase, alternating current, shall be applied for 1 minute between the exposed non-current-carrying parts and the current-carrying parts, with all current-carrying parts in the circuit. Test shall be performed at the maximum operating temperature reached in normal use.

4.4.3.3 Suction overflow. The suction system shall be operated until the cutoff device for the negative pressure operates. The device shall operate before any liquid from the suction bottle is drawn into the suction system.

4.4.3.4 Performance. The unit shall be operated in accordance with the manufacturer's instructions, and the following determinations shall be made:

Compressor capacity. (3.3.3)
Delivery at pressure end. (3.3.3.1)
Intake at suction end. (3.3.3.2)

The unit shall be completely assembled during the test. The suction bottle and the suction and pressure filter jars or glass shall be subjected to a min. pressure of 20 p.s.i.g. and a minimum vacuum of 20 inches of mercury, as applicable, for one minute without failure. Test gages shall have an accuracy of ± 1 percent.

5. PREPARATION FOR DELIVERY.

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.2).

5.1.1 Unit of issue. One suction and pressure apparatus, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. Each unit shall be disassembled as far as is practicable and packaged as follows:

5.1.2.1.1 Disassembled parts. Disassembled parts shall be adequately protected by wrapping with flexible corrugated fiberboard. Parts so wrapped shall then be packaged in a nailed wood box of suitable size and design. The box shall contain sufficient cushioning material (corrugated pads, air cells, or cellulose wadding) to insure against shifting and breakage.

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5.1.2.1.2 Motor and pumps. If the motor and pumps are flexibly mounted (floating type), they shall be adequately blocked or supported and secured with steel bands where practicable, in a manner to prevent movement of motor or pump while in transit or storage. A tag shall be attached to the cabinet directing the removal of all shipping straps and blocks prior to use of equipment.

5.1.2.1.3 Cabinet. Doors and drawers of the cabinet shall be secured to prevent accidental opening while in transit or storage. The cabinet shall then be wrapped with kraft paper of suitable weight and padded with corrugated fiberboard or with excelsior pads. Wrapping and padding shall be firmly secured with kraft gummed paper tape, or with cotton twine or tape, or with any combination of the three. All work shall be accomplished in a manner to prevent scratching or marring of finished surfaces or damage to the cabinet.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which shall afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. The cabinet, and the box containing disassembled components, shall be packed in an exterior container designed for type 3 load and constructed in accordance with Federal Specification PPP-B-621, Class 2. Box and cabinet shall be adequately secured within the exterior container by means of bolts, wood braces, or other interior fittings in such a manner as to insure against shifting, damage, or marring of surfaces. All box panels, except the base, shall be lined with waterproof barrier material conforming to PPP-B-1055, Class M-1 in such a manner as to prevent direct entry of water. When gross weight of the exterior container exceeds 250 pounds, exterior container design shall include skids of not less than 1-1/2 inches in thickness; applied crosswise of the base, so positioned as to facilitate handling with a forklift truck and to maintain unit in an upright position. The cabinet shall be blocked or braced in such a manner that the weight shall not be carried directly by the casters, and any protruding parts of the cabinet shall not contact the inner walls of the container. Closure shall comply with appendix of box specification. Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

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5.2.2.1 Exterior container. The cabinet, and the box containing disassembled components, shall be packed in an exterior container designed for type 3 load and constructed in accordance with Federal Specification PPP-B-621, Class 1. Box and cabinet shall be adequately secured within the exterior container by means of bolts, wood braces, or other interior fittings in such a manner as to insure against shifting, damage, or marring of surfaces. When gross weight of the exterior container exceeds 250 pounds, exterior container design shall include skids of not less than 1-1/2 inches in thickness, applied crosswise of the base, so positioned as to facilitate handling with a forklift truck and to maintain unit in an upright position. The cabinet shall be blocked or braced in such a manner that the weight shall not be carried directly by the casters, and any protruding parts of the cabinet shall not contact the inner walls of the container. Closure shall comply with appendix of box specification. Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Interior packages and exterior container shall be marked in accordance with Federal Standard No. 123.

5.3.2 Military activities.

5.3.2.1 Interior packages. Interior packages shall be marked to show the contents contained therein.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Marking shall include arrows as specified in MIL-STD-129. Marking shall also include the words "Front" or "Open this Side" as appropriate in order to facilitate opening.

6. NOTES

6.1 Intended use. Explosion-proof suction and pressure treatment unit for use by ear, nose, and throat specialists.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packing (see 5.1 and 5.2).

6.3 This specification covers the following item shown in the Federal Supply Catalog

NSN	Item
6515-00-312-8200	Suction and Pressure Apparatus, Surgical Explosion-Proof, Single Compressor, Mobile, 110 volt, 60 cycle, AC.

6.4 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

6.5 The thumb cut-off described in 3.4 shall conform to the latest requirements for National Stock Number 6515-00-386-9788; the Yankauer cannula to 6515-00-721-9474; and the sinus cleanser unit to 6515-00-099-8132, 6515-00-099-8150 and 6515-00-386-6157.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1682

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

GG-S-001090A (DSA-DM)

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CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

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